1. Clinical Question & Core Hypothesis

Clinical Question

The Ontologist highlights that noninvasive immune readouts (Immunoscore/TILs) and tumor-derived signals (urine DNA methylation, ADNP mRNA in CTCs/ctRNA) each predict NMIBC outcomes, yet immune-rich tumors do not always benefit from immune-based therapies or intravesical BCG. Why does local immune infiltration sometimes fail to translate into therapeutic response?

Working Hypothesis

In NMIBC, the predictive value of Tumor-Infiltrating Lymphocyte (TIL) density for therapy_response is contingent on tumor-intrinsic ADNP expression. Specifically, high TIL density predicts favorable therapy_response to intravesical immunotherapy or systemic checkpoint blockade only when tumor ADNP expression is low; high ADNP expression drives chromatin remodeling that dampens antigen presentation or T cell effector function, offsetting the benefit of TILs. Therefore, a combined biomarker model using TIL density and ADNP status (with CD8A as a CD8+ T cell readout) will outperform TIL density alone in predicting therapy_response (endpoint). Secondary modifiers—urine DNA methylation signatures, CTC/ctRNA ADNP, HDL-C, HRR, and MSI—will refine this interaction and improve risk stratification by capturing tumor-epigenetic state and systemic immune context, but the primary mechanistic claim remains the TIL × ADNP interaction driving response or resistance.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	ADNP, CD8A
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

ADNP: ADNP mRNA detected in liquid biopsy/CTCs as a tumor-derived transcript with chromatin-regulatory potential; and (c) Immunoscore/CD8+ T cell density (operationalized here as TIL density) representing the local adaptive immune contexture.

CD8A: Selected as a core mechanistic driver from Tier 1 knowledge graph analysis.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	urine DNA methylation signatures	Supporting modifier from knowledge graph

Secondary	CTCs / ctRNA	Supporting modifier from knowledge graph
Secondary	HDL-C	Supporting modifier from knowledge graph
Secondary	HRR (hemoglobin-to-RDW ratio)	Supporting modifier from knowledge graph
Secondary	MSI	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: clinical association studies assessing TIL density and ADNP expression in tumor samples, with therapy_response as the outcome; test for interaction between TIL and ADNP;

Prediction 2: measurement of CD8A alongside ADNP to confirm the immune-context;

Prediction 3: noninvasive correlates using urine DNA methylation panels and ctRNA for ADNP to monitor tumor state and link to therapy_response;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Gene expression does not equal protein activity; phospho-protein or functional assays are needed for confirmation.

Alternative Directions

- urine DNA methylation signatures: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- CTCs / ctRNA: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.